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1. Introduction

1.1 Background and Problem Statement

Global migration has emerged as a significant and more intricate phenomena of the 21st century, with around 281 million international migrants documented by 2020. The extensive magnitude of this human mobility poses considerable issues for governments and international organizations, which necessitate timely and precise data to guide policy, distribute resources, and strategize for the future. Nonetheless, a significant issue persists: conventional techniques for analysing migration data frequently fail to adapt to the dynamic and non-linear characteristics of these global movements. These established methodologies, however fundamental, may exhibit constraints in their predictive efficacy, resulting in a disparity between raw data and the actionable insights necessary for effective decision-making. This study is driven by the rise of machine learning (ML) as a revolutionary opportunity to address this gap, providing a means to go from descriptive statistics to more robust, predictive, and insightful analytical frameworks.

1.2 Research Questions

To address the problem statement, this dissertation seeks to answer the following fundamental research questions:

- **RQ1:** Which countries consistently receive the largest number of international migrants?
- **RQ2:** What are the global patterns of gender distribution among migrants, and how have these trends evolved over time?

1.3 Aim and Objectives

This study aims to examine the UN International Migrant Stock dataset through a machine learning framework. The primary aims are to systematically determine the leading destination

and origin nations for international migrants annually from 1990 to 2024. The study will simultaneously examine the global gender distribution of migrants, categorizing countries to ascertain if their migrant populations are primarily male or female.

1.4 Scope

The scope of this research is explicitly determined by the dataset and the analytical techniques employed. The research solely employs the United Nations Department of Economic and Social Affairs (UN DESA) International Migrant Stock dataset, spanning from 1990 to 2024. The analytical emphasis is on utilizing machine learning models for classification to determine leading origin and destination nations and to examine gender trends, as well as for time-series forecasting to predict future migrant stock levels. This research does not seek to explore the qualitative or profound socio-economic root reasons of single migration events; it concentrates on the quantitative examination of macro-level migration patterns as indicated by official data.

1.5 Significance of the Study

This research is significant for two reasons: it contributes to academic knowledge and has practical applications. The paper identifies a notable gap in the literature by developing and evaluating a comprehensive machine learning pipeline for a worldwide migration dataset. It seeks to transcend the regionally or thematically focused methodologies of numerous previous research, offering a holistic framework for identifying and predicting overall migrant stock. The findings may provide significant insights for politicians, international organizations, and NGOs. This research can offer a more detailed data base for strategic planning by identifying essential origin and destination hubs and examining changing gender dynamics. This data-driven knowledge can facilitate more efficient resource allocation, guide targeted social policies, and improve long-term foresight in an era of escalating human mobility.

1.6 Dissertation Structure

The subsequent sections of this dissertation are structured into five chapters. Chapter 2 offers a comprehensive evaluation of the pertinent literature regarding migration analysis and machine learning, ultimately pinpointing the precise research need. Chapter 3 outlines the research approach, specifying the data source, preprocessing techniques, and the array of machine learning models utilized for categorization and forecasting. Chapter 4 delineates the results of the analysis, encompassing model performance measures and principal findings. Chapter 5 examines the ramifications of these findings, contextualizing them within the research questions and the wider body of literature. Chapter 6 ultimately finishes the dissertation by summarizing the principal contributions, recognizing the study's limits, and providing recommendations for further research.

2. Literature Review

2.1 Introduction

People moving across international borders is a key feature of the current world that is becoming more globalized. A complicated mix of economic, social, and political pressures is causing this to happen, and it has big effects for policy, business, and the growth of society. As more people move between countries, it is becoming increasingly important for governments, non-governmental organizations (NGOs), and businesses to use data to better understand the patterns of international migration. Some evidence suggests that old analytical approaches, while important, may not be able to fully capture the changing nature of modern migratory movements, which means we need better analytical tools [7].

This chapter critically reviews existing literature combining migration studies and predictive analytics, positioning the current study within established research and providing theoretical foundation for the methodology. The paper will explore the progress of migration analysis, ranging from historical descriptive techniques to present machine learning applications. It will look closely at the Random Forest, XGBoost, Support Vector Machines (SVM), and ARIMA models that are important to this dissertation by talking about how they are used, how well they work, and what their limits are based on studies that are relevant.

A main part of this study, which follows standard academic practice, will be a thorough look at a few important research that have used data-driven methods to look at migration. This review will slowly create a case for the specific method used in this dissertation by looking at the methodologies, metrics, and flaws in important publications by authors like Carammia et al. [5], Halim and Seetahul [24], and Zagatti et al. [23]. The study will end with the discovery of a specific research gap: the lack of a single, worldwide framework that uses machine learning to classify and predict the number of international migrants.

The review is set up in a way that leads from a general understanding of the topic to this specific research gap. The next parts will look at how migration analytics can help with decision-making, how the methods have changed from statistical to machine learning approaches, a critical review of relevant algorithms and key case studies, and finally, a summary of the ongoing problems in the field, which directly relates to the contribution of this research.

2.2 The Strategic Importance of Migration Analytics

In today's world, the strategic analysis of migration data has gone beyond being just an academic exercise to become an important part of good governance, economic strategy, and humanitarian action [4]. As more people move around and it gets more complicated, we need strong analytical frameworks that can give a lot of different people useful information. Evidence demonstrates that comprehensive, data-driven approaches are essential for managing migration's multifaceted impacts, enabling decision-makers to shift from reactive to proactive, evidence-based planning [8].

One of the main jobs of migration analytics is to base public and political discussions on real data, which can help fight the myths and misunderstandings that can lead to policies that don't work or are wrong [3]. The International Organization for Migration (IOM) supports using data for "action, insight, and foresight." This shows a strategic change toward employing analytics not only to understand the present but also to predict future needs and possibilities [4]. This perspective emphasizes the importance of evidence-based policy development, where comprehensive data informs everything from initial problem identification to the rigorous monitoring and evaluation of implemented solutions [8]. It seems that having detailed, broken-down data on migrants' characteristics beyond just numbers is very important for creating targeted and successful interventions [3].

From a commercial and economic intelligence point of view, migration analytics are important for planning at the national and international levels. To figure out how migrants affect public budgets, labor markets, and job creation in host countries, we need to know how many migrants are coming and going and how well they are integrating [9]. For example, research by the Organisation for Economic Co-operation and Development (OECD) shows that immigrants pay more in taxes and social contributions than they get in public benefits. This information directly affects fiscal policy and public discourse [9]. Also, these analytics are used in macroeconomic models to predict how they will affect GDP growth and to figure out how long social security systems will be able to pay out benefits, especially in countries with older populations [10].

The use of predictive analytics is especially groundbreaking in the field of humanitarian assistance. In times of crisis caused by conflict or climate change, timely migration data is essential for providing vital help and saving lives [4]. More and more, organizations are using machine learning to fuel forecasting algorithms that can anticipate when refugees will arrive and when people will move within a country. This lets them plan ahead for how to use resources and staff [11]. The World Bank's AI-powered model in Uganda is a well-known example. It can predict how many refugees will arrive up to six months in advance. This foresight lets the government and its partners build water stations, add to health facilities, and get classrooms ready before a lot of people come, which makes the community stronger and less tense overall [11]. This move toward taking action based on predictions, thanks to predictive analytics, is a big step forward in strategic resource allocation and humanitarian logistics.

2.3 Evolution of Migration Analysis: From Statistical Description to Predictive Analytics

Migration research has changed a lot in terms of how it is done. It used to rely mostly on descriptive statistical approaches, but now it is moving toward using computational science to make predictions. The main reason for this change is that old methods can't fully capture how complex and changing human mobility is nowadays. At the same time, machine learning (ML) [6], [7] is becoming more powerful.

2.3.1 Traditional Demographic and Statistical Methods

In the past, studying migration was based on well-known demographic and statistical methods. These strategies were very important for figuring out how many people moved in the past and why they did. Some of the most common methods were the cohort-component method, which makes predictions about future population numbers based on assumptions about fertility, mortality, and migration [16], and theoretical frameworks like Lee's "push-pull" model, which sorts the reasons why people leave or move to a new place [18]. Gravity models were also used a lot to explain migratory flows depending on the size and distance between the places of origin and destination [17]. These methods are useful for giving descriptive and explanatory information, but they typically have problems in today's data-rich world.

2.3.2 What are the problems with traditional methods?

Traditional approaches are very important, but they have some problems that make them less useful for modeling complicated, modern migratory patterns. One of the main problems with them is that they have trouble making strong causal ties and dealing with non-linear relationships. They are better at finding correlations than proving causality [7]. This often makes the migration process seem too simple, since simple push-pull frameworks may not fully explain how people's goals, abilities, and macro-structural changes interact, such as the "trapped populations" who want to move but don't have the resources [20].

Also, these methods are often limited by problems with the quality and timeliness of the data. Official statistics are reliable, but they often take a long time to be published, which means that decision-makers have to use old data [5]. There is also a lot of fragmentation in the worldwide data landscape, with countries using different definitions and not breaking down data by important demographics like age and gender enough, which makes it hard to do full and comparable research [3].

2.3.3 The Emergence of Machine Learning in Migration Analysis

These constraints have left a gap in the methods that has made it possible for machine learning and computational social science techniques to be used [21]. The fact that ML models are the best tools for dealing with the complexity, volume, and speed of modern migration data is what is driving this change in thinking. Several important benefits seem to support this change.

First, ML models are better at handling complicated, high-dimensional, and non-linear datasets. This lets us combine a wide range of socio-economic, political, and environmental factors that affect migration [7]. Second, ML algorithms can change on their own, but static statistical models need to be recalibrated by hand. ML algorithms can keep becoming better at predicting things as new data comes in [7]. Finally, one of the most important things about ML is that it can use new "big data" from digital traces, including cell phone signals, satellite images, and internet search queries, to "nowcast" current migration trends in almost real time [23]. This feature directly addresses the important problem of data latency in official statistics, which could lead to policy and corporate information that is more immediate and responsive. This change is a major step forward because it allows for a more detailed and forward-looking look at global mobility.

2.4 A Critical Review of Machine Learning Models That Are Relevant

Choosing the right analytical model depends on the research objective, the type of data, and the right balance between how well it predicts and how easy it is to understand. This study uses a set of machine learning models, including Random Forest, XGBoost, and Support Vector Machines for classification and regression, as well as ARIMA for forecasting time series. The review that follows looks at how well these models have worked in the past and how they can be used in situations that are important to migration research.

2.4.1 Random Forest

Random Forest is a flexible ensemble learning method that is known for being quite accurate at making predictions and not being too sensitive to overfitting [27]. During training, the method builds a lot of decision trees and then gives the mode of the classes (for classification) or the mean prediction (for regression) of the individual trees. This method has worked well in a number of social science fields. For example, research has shown that Random Forest models can find subtle connections between factors that affect migration decisions better than classic statistical methods like logistic regression [27]. It is especially good for classifying nations based on their migration profiles and figuring out the main reasons behind those classifications because it can handle complicated, non-linear relationships and show how important different variables are.

2.4.2 Xtreme Gradient Boosting (XGBoost)

Xtreme Gradient Boosting (XGBoost) is a very effective and scalable version of gradient boosting that has always produced the best results in many machine learning situations. The algorithm creates a strong prediction model by adding new decision trees one at a time, each one fixing the mistakes of the one before it. Its architecture has advanced regularization methods that assist keep the model from overfitting and make it better at generalizing to new data. According to the research, XGBoost is quite good at predicting demographics, especially when it uses data from several different sources. One study that combined traditional census data with mobile phone data employed an XGBoost ensemble to predict sub-national population statistics with a R^2 value of 0.9984, which shows how well it works [24]. XGBoost is a great choice for improving the classification performance in this study because it has been shown to work well with large, structured datasets.

2.4.3 Support Vector Machines (SVM)

Support Vector Machines (SVM) are a type of supervised learning model that works very well for classification problems. The main idea behind an SVM is to discover the best hyperplane that separates data points of distinct classes in a space with a lot of dimensions. SVMs can handle non-linear correlations well by using the "kernel trick" to translate data into feature spaces where linear separation is possible. This means that the model can handle noisy data and work well even when there are a lot of dimensions. SVMs have been used successfully to guess how people will move about in certain cities [26]. It is a good strategy for the binary classification tasks in this study, including figuring out if a country's migrant population is mostly male or female, because it makes obvious and distinct lines between groups.

2.4.4 The Autoregressive Integrated Moving Average (ARIMA)

The ARIMA model is a basic statistical tool for looking at and predicting time-series data. It uses past values of the variable (autoregression) and past forecast mistakes (moving average) to make future predictions [19]. This is how it captures temporal dependencies in a dataset, like trends and seasonality. ARIMA has been used for a long time in demographic settings, such as predicting population growth and migration. The literature also constantly points out a major problem: migration is commonly seen as the most unpredictable part of population development, which makes it hard to model with approaches that depend on previous patterns. Researchers who use ARIMA to predict migration, like Tolesh and Biloshchytska in Kazakhstan [19], often talk about problems with data quality and the model's ability to handle abrupt shocks. ARIMA is a useful statistical starting point for our forecasting goal, but its known problems in the unstable field of migration show why we need the more flexible, non-linear machine learning models used in other parts of this work.

2.5 Related Work and Case Studies in ML-Driven Migration Analysis

A growing corpus of empirical studies is backing up the theoretical benefits of using machine learning for migration research. These case studies show how ML may be used in real life to provide prediction insights and analysis tools for certain migration events. But a close look at their breadth and methods also shows the setting of the current investigation.

2.5.1 Forecasting Asylum-Related Flows in Europe (Carammia et al. [5])

The work of Carammia et al. is a big step forward in predicting migration. Their study was a direct response to the "refugee crisis" in Europe from 2015 to 2016, which showed that standard forecasting methodologies were fundamentally flawed and left governments unprepared for the rapid rise in asylum applications. The study used machine learning algorithms to create a data-driven, adaptive system that makes predictions that are more timely and detailed.

A major strength of their method was how well they combined data from different sources. The model combined typical administrative data with new, non-traditional inputs on a large scale. These included geolocated conflict occurrences, Google Trends data on internet searches for migration-related phrases like "visa" and "refugee," and Frontex data on irregular border crossings. The technology was able to predict asylum flows between certain countries with a high degree of accuracy up to four weeks in advance. This work is important because it shows how powerful ML may be for predicting forced displacement. However, it is different from other studies in that it focuses on asylum seekers and is mostly about the European Union. In contrast, our study tries to estimate the overall number of international migrants around the world, which includes a wider spectrum of migration drivers than only those that apply to refugee situations.

2.5.2 Identifying Gender-Specific Migration Drivers (Halim and Seetahul [24])

Halim and Seetahul's research makes a strong case for the need for a gender-disaggregated examination of migration. Their study used a data-driven method with supervised learning models to find the different things that affect men's and women's decisions to move. Their

results support a key idea in our research: that migration patterns and their causes are not the same for everyone, and that looking at them through a gendered lens might disclose important information that would be hidden in aggregated data.

But Halim and Seetahul's method is different from ours in two important ways. First, their study only looked at one country (Indonesia), while ours looks at the whole world. Second, and more importantly, their study was based on survey data that showed

the desire to move, not actual moves that have already happened. Intention is useful, but it doesn't always lead to action. Our study uses the UN's International Migrant Stock dataset to look at the amount of foreign-born people already living in countries. This means that we are more interested in the results of migration processes than in people's intentions.

2.5.3 "Nowcasting" Migrant Stocks with Social Media Data (Zagatti et al. [23])

A study by Zagatti et al. at the RAND Corporation created a method to "nowcast"—or give real-time estimates of—migrant stocks. This was done to deal with the long time gaps that are commonly found in official statistics. They used a Bayesian model that cleverly merged official migration statistics with real-time digital trace data from the Facebook Marketing API. They used users' "Lived in" status as a stand-in for their country of origin. This method worked especially well for making quick estimates for events that happened quickly, like the COVID-19 pandemic, when official data wasn't available yet.

The study was very careful to point out the problems with using commercially generated data, even though it was a new way to look at timeliness. Some of these problems are that social media user bases have built-in selection biases, platforms don't always make their rules clear, and algorithm updates can cause rapid, inexplicable changes in the data. This important self-evaluation strongly supports our study's use of the UN DESA dataset. Our chosen dataset may take longer to publish, but it is more methodologically clear, consistent over time, and covers the whole world, which are all important for creating a strong and generalizable categorization and forecasting system.

These case studies show a definite trend toward using specialized ML models for some aspects of migration. They are strong, but they tend to just look at certain areas, categories of migrants, or data sources. This specialization makes it possible to take a more comprehensive approach, which will be explained in the next part.

2.6 Challenges and Identification of the Research Gap

Even if the case studies show that there have been big improvements, the larger body of literature shows that modeling global migration is still a very difficult task. There are still some big problems that need to be solved, and together they show the exact research gap that this dissertation will fill.

First, there are several "push" and "pull" forces that work together in a very complicated and changing way to cause migration [18]. The choice to move is a complicated mix of people's goals and capacities, which simple models don't always show. There are "trapped populations,"

or people who want to move but don't have the money to do so. This makes it much harder to see how bad conditions lead to migration flows [20].

Second, problems with data availability, quality, and standardization continue to make it hard to do a full analysis [3]. Official statistics can take a long time to come out [5], and the definitions of data can be different from country to country, which makes it hard to make strong comparisons between countries. New "big data" sources are timely, but they also have problems with accessibility, openness, and possible bias [23].

Finally, because global events are so unpredictable, such wars and natural disasters caused by climate change, it's very hard to predict migration using models that assume stability over time [30]. These challenges mean that while specialized ML applications have proven highly productive, they are frequently confined in scope.

The assessment of linked studies above shows that there is a clear and important gap in the present research. The case studies are very useful, but they mostly look at specific migratory events, such asylum seekers [5], gender-specific factors in one region [24], or real-time estimates for certain routes using proprietary data [23]. While existing studies have applied machine learning to specific regional contexts or migration types, no study has systematically evaluated multiple ML algorithms on the complete global UN DESA migration stock dataset while simultaneously addressing both regression (stock prediction) and classification (destination popularity, gender patterns) tasks through gender-disaggregated modeling. This study addresses this gap by implementing a dual-model machine learning approach that separately predicts male and female migrant stocks before combining them for total migration forecasts. This gender-disaggregated methodology is applied systematically across multiple algorithms (Random Forest, XGBoost, SVR) using the complete UN DESA global dataset to simultaneously achieve regression prediction and classification tasks.

2.7 Conclusion

This review has shown how important it is to have strong migration analytics to help make decisions about policy, economic planning, and humanitarian response. It has shown how migration studies have changed over time, from using simple descriptive statistics to using machine learning to make predictions. It argues that ML models are superior at dealing with the complexity, size, and speed of modern global data. The review has shown that Random Forest, XGBoost, SVM, and ARIMA models work well in certain analytical situations and has also pointed out their benefits and weaknesses.

A close look at important case studies shows that there is a significant trend toward using machine learning for specialized migration analysis. But this same study has also shown a big research gap: there is no complete, global framework that combines both the classification and the forecasting of the total number of international migrants. This literature review sets the stage for the current investigation by giving it a strong reason to exist. The next chapters will explain how this research was done to fill this gap and show what was found.

3. Methodology

This chapter covers the methodical, systematic technical process employed to assess the migration data and construct the predictive models.

Rationale for a Regression-Based Approach

A strategic decision was made initially to treat the core analytical work as a regression problem, despite the presence of classification elements in the research questions. The principal reason for this is that the target variables—the counts of male and female migrants—are fundamentally numerical. Although statistically optimal models for count data, such as Generalized Linear Models (GLMs), were considered, the choice was taken to utilize tree-based regressors like Random Forest. This decision was influenced by their proven higher predictive accuracy, resilience in managing intricate, non-linear interactions, and relative ease of the execution in a business analytics framework.

This regression-first approach facilitated a flexible, two-phase strategy to fulfill the research questions. The regression models were initially trained to generate direct numerical projections of migrant stock, crucial for forecasting purposes. Secondly, to address the classification-based research questions, these continuous numerical outputs were transformed into discrete categories (e.g., 'Popular Destination' versus 'Not Popular'). This methodology allowed a single set of robust regression models to effectively address both the quantitative forecasting and qualitative categorization goals of the project.

3.1 Data Source and Description

This analysis is based on an extensive dataset of global migration statistics obtained from the United Nations Department of Economic and Social Affairs (UN DESA). The data is initially organized in a broad style, with each row representing a distinct migration corridor by indicating a country of origin and a country of destination.

The columns denote migrant stock levels at different intervals from 1990 to 2024. The data is further categorized by gender, featuring distinct columns for the total migrant population and for male and female migrant numbers for each corresponding year. This initial structure, although thorough, is not immediately applicable for machine learning and necessitates considerable modification for effective analysis, as will be elaborated in the subsequent section.

3.2 Data Preprocessing and Transformation

The complete raw data necessitated extensive preparation to be appropriate for machine learning. This phase concentrated on sanitizing the dataset to guarantee analytical integrity and restructuring its framework for optimal modeling.

A data cleaning method was executed to narrow the focus of the analysis to sovereign, bilateral migration. An exhaustive exclusion list was created to eliminate rows that include aggregate geographical regions (e.g., 'AFRICA'), economic classifications (e.g., 'High-income

countries'), and various non-sovereign territories. This stage was essential for guaranteeing the quality and usefulness of the data, enabling the future study to concentrate specifically on the interactions among individual nations.

Secondly, the dataset experienced a structural alteration. The original wide format was converted into a long, or 'tidy', format, which is standard for rigorous data analysis. This intermediary step enabled the year and migrant gender to be regarded as separate variables. The data was subsequently rotated to establish the final, structured analytical foundation. In this final style, each row denotes a distinct observation for a unique migration corridor (an origin-destination pair) in a certain year, with designated columns for male, female, and total migrants.

3.2.1 - Data Quality Considerations

Several data quality issues were addressed during preprocessing. Missing values in migrant counts were handled by imputing zeros. The dataset spans periods of significant geopolitical changes (e.g., Soviet Union dissolution, Yugoslav wars), requiring careful treatment of successor states. The UN DESA methodology maintains consistent definitions across the time period, though data reliability varies by country based on national statistical capacity.

3.3 Feature Engineering

After the data transformation, the following stage was feature engineering, during which a more comprehensive array of predictor variables was developed to augment the models' capacity to identify underlying patterns. This procedure concentrated on producing historical, trend-oriented features from the structured dataset.

Several lag features were generated to furnish the models with historical background for each migration route. Lagged values for male, female, and total migrant counts were produced for one-year and two-year intervals, as migration patterns typically exhibit short-term persistence while longer lags may introduce noise from changing economic and political conditions. This approach captures recent trends without incorporating outdated historical context. These attributes function as a temporary "memory," enabling the model to account for the latest trends during prediction.

In addition to historical values, a set of features was developed to encapsulate the dynamics and composition of migrant flows. These growth rate features capture migration momentum and trajectory changes, providing dynamic indicators beyond static stock levels. These variables furnish the model with essential insights about the momentum and internal patterns of migration along a certain path, which might be more indicative than mere absolute figures.

Ultimately, to directly tackle the research questions, a set of binary classification variables was developed. These variables classify a destination country or a certain route as "popular" (ranked in the top percentile of migrant stock for a given year) and indicate whether a corridor was predominantly male or female. These engineered flags function as the definitive labels for the ensuing evaluation of classification performance.

3.4 Model Input Preparation

Post feature engineering, the dataset was prepared for machine learning models through critical steps: dividing the data into training and testing sets, encoding categorical variables, and normalizing numerical features.

A crucial step in this procedure was the chronological train-test division. To accurately replicate a forecasting scenario, the data was not randomly partitioned. All data from 1990 to 2020 was designated for the training set, utilized for instructing the models. The data for 2024 was retained as an entirely unobserved test set. This methodology guarantees a thorough assessment of the models' capacity to generalize and forecast future results based on historical trends.

Then, two preprocessing approaches were implemented. The high-cardinality categorical variables, dedicated for origin and destination country names of the migration corridor, were encoded utilizing a Target Encoder. This method transforms nation names into numerical values according to their association with the target variable, thereby embedding predictive information within the features. Finally, all numerical attributes in the dataset were standardized utilizing a Standard Scaler. This conventional procedure guarantees that characteristics with greater magnitudes do not significantly affect the model's training process.

3.5 Modeling and Training Strategy

The modeling strategy involved a multi-stage process aimed at initially identifying the most effective algorithm and subsequently enhancing its performance through rigorous tuning.

An initial baseline comparison was performed to determine the most appropriate model family for this issue. Three distinct algorithms - Random Forest, XGBoost, and Support Vector Machine (SVR) - were trained and assessed. A baseline comparison was performed between Random Forest, XGBoost, and Support Vector Machine (SVR) algorithms. Random Forest demonstrated superior performance (detailed results in Section 4.x) and was selected as the primary algorithm for the final models.

A crucial decision was subsequently made to create two distinct, specialized models instead of a singular monolithic model. A Random Forest model was developed just to forecast male migrant stock, while a separate model was created for forecasting female migrant stock. The ultimate forecast for total migrants was subsequently obtained by aggregating the results of these two specialized models. This methodology enables the framework to identify the possibly divergent patterns and determinants of male and female migration, resulting in a more refined and precise overall prediction.

Ultimately, both the male and female Random Forest models underwent a rigorous hyperparameter tuning process in order to determine their best configurations. This was carried out with a Randomized Search methodology in conjunction with TimeSeriesSplit cross-validation. The optimal hyperparameters and cross-validation performance for both models are detailed in Section 4.x. This cross-validation method was crucial for guaranteeing a

chronologically valid and robust tuning procedure, since it avoids data leaking from future periods into the training folds, thereby offering a more accurate assessment of the models' real-world predicting efficacy.

3.6 Model Evaluation Metrics

A comprehensive set of industry-standard metrics were used to meticulously assess the models' performance. The assessment was performed from two viewpoints: the accuracy of the regression task (forecasting the migrant count) and the effectiveness of the classification task (addressing the research questions).

3.6.1 Evaluating Regression Performance

The core prediction accuracy of the Random Forest models was evaluated using the following regression metrics:

Mean Absolute Error (MAE): This metric offers a clear and interpretable measurement of the average prediction error, expressed in the same unit as the target variable (number of migrants). It directly refers to the business question: "What is the average deviation of our forecast?"

Root Mean Squared Error (RMSE): This measure parallels Mean Absolute Error (MAE) but gives greater significance to larger errors. It is particularly susceptible to substantial errors in forecasting that may have major implications for planning and allocation of resources.

R-squared (R^2): This denotes the fraction of variance in the migrant stock elucidated by the model's attributes. An elevated R^2 value indicates superior model fit and enhanced explanatory capability.

3.6.2 Evaluating Classification Performance

In order to evaluate the model's effectiveness in dealing with the categorical research questions, the regression outputs were first used to generate the binary classification indicators (e.g., Is_Popular_Destination). The accuracy of these derived classifications was further assessed using the following:

Confusion Matrix: A tabular structure used to demonstrate the validity of classification, showing the amount of accurate and incorrect forecasts for every class.

Precision, Recall, and F1-Score: These measures offer a more refined perspective on classification accuracy. Precision quantifies the accuracy of positive predictions (of all routes designated as 'popular', how many were genuinely so?), whereas Recall assesses the model's capacity to identify all true positive instances (of all the authentically 'popular' routes, how many did we detect?). The F1-Score offers a singular metric that balances precision and recall.

4. Results and Analysis

This chapter highlights the factual outcomes of the study, beginning with the descriptive insights derived from the Exploratory Data Analysis (EDA) and moving to the performance evaluation of the predictive models. Every result is presented and subsequently analyzed from an analytical viewpoint.

4.1 Exploratory Data Analysis: The Global Migration Landscape

The study begins with an examination of the major patterns in international migrant stock during the 34-year span of the dataset. The figure below depicts the rise in total, male, and female migrant populations from 1990 to 2024.

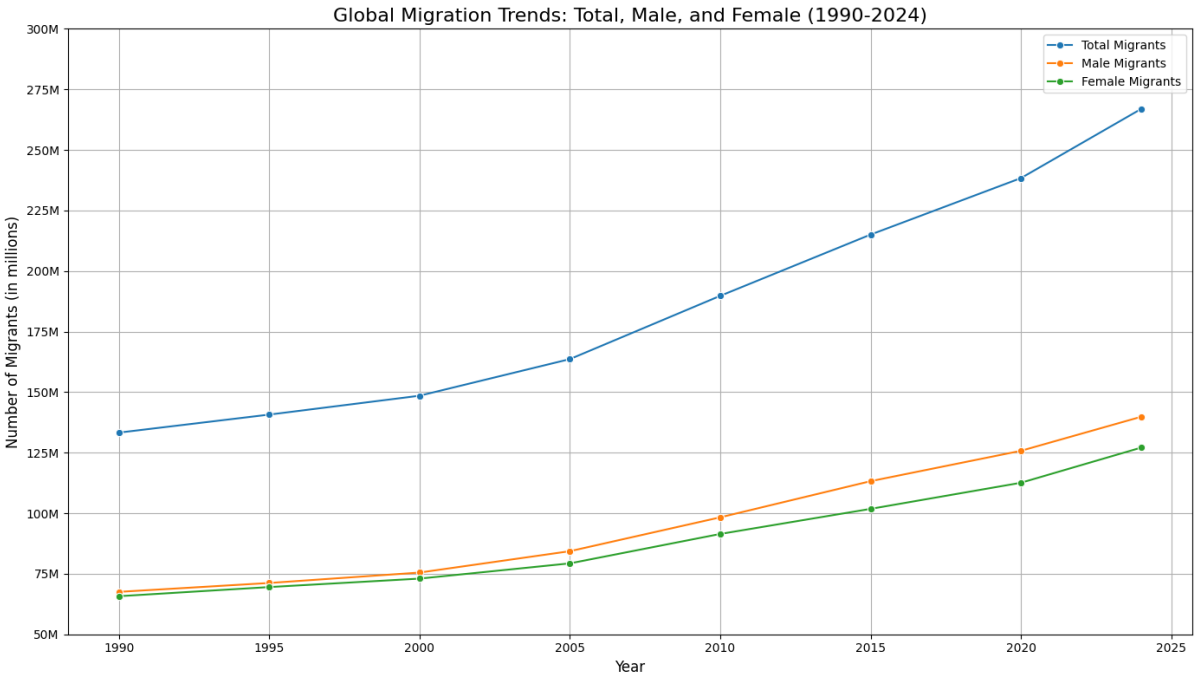


Figure 1. Global Migration Trends by Gender (1990-2024)

The most significant observation from this visualization is the substantial and continuous increase in the total count of international migrants. The study shows that the worldwide migrant stock has essentially doubled, rising from roughly 153 million in 1990 to more than 300 million by 2024. This steady, nearly linear upward trend highlights the increasing significance of migration as a reliable and impactful element in global economic strategy, international labor markets, and long-term demographic projections.

A detailed examination of the gender-disaggregated data uncovers another persistent trend. During the whole duration, the quantity of male migrants has continuously surpassed that of female migrants. Although both populations have expanded simultaneously, the absolute gap between them seems to have increased over time. This continuous gender disparity is a significant learning, indicating possible consequences for creating policies concerning social integration, healthcare, and the economic evaluation of remittance flows, as these could show varying patterns according to gender.

4.2 Identifying Key Global Migration Hubs

A comprehensive look is required to determine the specific nations which act as the primary hubs for international migration, beyond the general worldwide trend. This investigation examines the total migrant stock from 1990 to 2024 to determine the top origin and destination countries.

The data reveals a significant concentration of migrants in just a few of destination nations. The figure below indicates these nations regularly drive worldwide talent and labor.

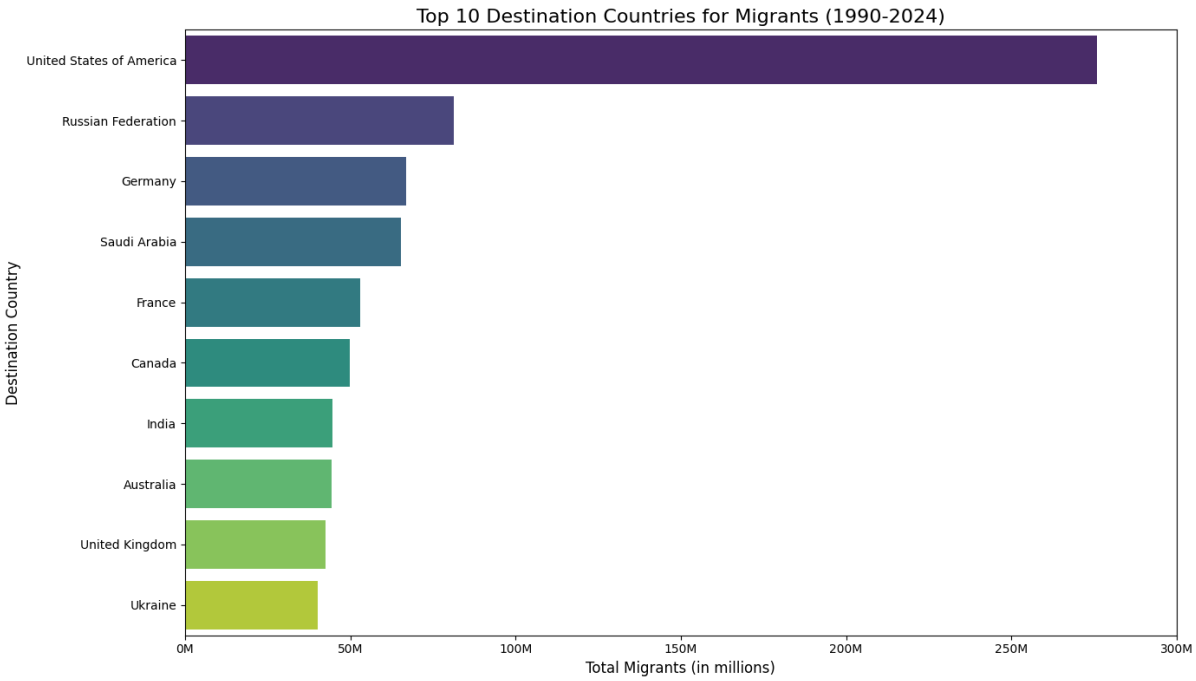


Figure 2: Top 10 Destination Countries for Migrants (1990-2024)

The United States emerges as the primary destination, with a total migrant stock exceeding 250 million throughout this period of time, establishing it as the largest recipient of international migrants by a substantial amount. Germany ranks next, holding a cumulative stock above 400 million, so confirming its status as a pivotal migration center, likely influenced by economic prospects inside the European Union and its immigration policy. The inclusion of countries like Saudi Arabia, the United Arab Emirates, the United Kingdom, and Canada in the top tier emphasizes that primary destinations are mainly prosperous nations. This concentration greatly influences labor market analysis, strategic workforce planning, and consumer demand forecasts in these particular economies from a business analytics viewpoint.

The data shows a significant proportion of countries of origin in the migration pattern. The figure below shows the top 10 countries of origin with the highest number of migrants.

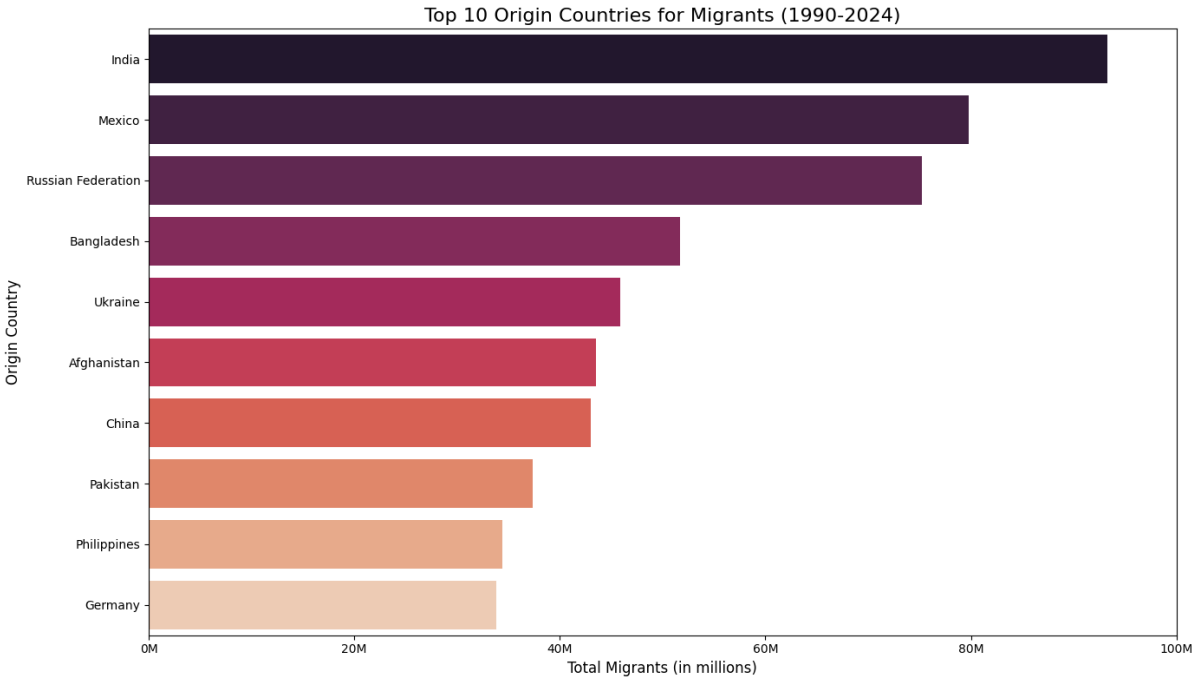


Figure 3: Top 10 Origin Countries for Migrants (1990-2024)

India is the largest nation of origin, with a total diaspora exceeding 400 million individuals staying abroad throughout the examined period. Mexico and the Russian Federation evidently showcase cumulative emigrant populations above 250 million. The list of primary origin countries has greater diversity than that of the destination countries, comprising major developing economies and populated countries from Asia and Eastern Europe.

This finding is of great significance from a business intelligence standpoint. It acknowledges countries with extensive and geographically dispersed diasporas, which provide major opportunities for international remittance services, telecommunications, and targeted trade. The study demonstrates the significant economic and social value of immigrant engagement for policymakers in these origin nations. As a whole, this investigations reveal a global migration framework that involves distinct and stable roadways linking particular migrants from and to countries.

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